Venkat Dinesh Indupuri
Software Engineer(AI/ML)
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SUMMARY:

AI/ML Engineer with 3+ years of experience building and deploying machine learning, NLP, and Generative AI solutions in enterprise environments. Technical expertise in Python, Azure OpenAI, Databricks, Spark, and MLOps pipelines (MLflow, CI/CD). Delivered risk analytics, LLM-based document intelligence, and explainable AI solutions that improved model accuracy by 20% and reduced operational cycles by 40% for regulated US enterprises.

TECHNICAL SKILLS:

- **Machine Learning & AI:** XGBoost, LightGBM, Random Forest, Logistic Regression, Time-Series (ARIMA, Prophet, LSTM), Feature Engineering, Recommendation Systems.
- **Generative AI & NLP:** Azure OpenAI, GPT Models, LLM Document Processing, Prompt Engineering, Text Classification, Sentiment Analysis (spaCy, NLTK, Gensim).
- **Programming & Data:** Python, Java, SQL, PySpark, REST APIs, Scikit-learn, TensorFlow, PyTorch.
- **Big Data & Streaming:** Apache Spark (Batch & Streaming), Kafka, Hadoop, Hive.
- **Cloud & MLOps:** Microsoft Azure, AWS (S3/EC2), Azure Databricks, MLflow, Docker, CI/CD Pipelines, Jenkins.
- **Model Governance:** SHAP, LIME, Model Monitoring, Compliance-Ready ML.

PROFESSIONAL EXPERIENCE:

Citigroup — **Jul 2025 - Current**

Role: Software Engineer(AI/ML)

Responsibilities:

- Engineered and deployed machine learning models (XGBoost, LightGBM) for risk profiling, improving model accuracy by 20% using historical transaction data.
- Automated 90% of manual classification tasks by orchestrating LLM-based document intelligence via Azure OpenAI and Databricks.
- Spearheaded NLP pipelines leveraging transformer-based models for policy analysis while ensuring 100% regulatory compliance.
- Designed AutoML workflows to streamline feature engineering, reducing experimentation cycles by 40%.
- Improved risk exposure forecasting by 15% using Prophet and LSTM models to predict claim trends for operational planning.
- Engineered near real-time data pipelines using Spark, Kafka, and REST APIs to support high volume transactional processing.
- Reduced deployment failures by 25% through automated MLOps pipelines using MLflow integrated with CI/CD workflows.
- Applied model explainability techniques (SHAP, LIME) to enhance transparency and audit readiness for production ML models.

Subex, India **Jun 2021 – Nov 2023**

Role: Software Engineer

Responsibilities:

- Designed and implemented machine learning solutions for customer segmentation, churn prediction, and campaign performance analysis using Logistic Regression, Random Forest, XGBoost, and LightGBM, supporting data-driven marketing and customer retention initiatives.
- Engineered predictive analytics and time-series forecasting models using ARIMA and LSTM to analyze global sales trends, which enhanced forecasting precision by 18% and optimized inventory planning.
- Applied Natural Language Processing (NLP) techniques using spaCy, NLTK, and Gensim to analyze large volumes of customer feedback and interaction data, generating sentiment and topic insights for business teams.
- Developed recommendation and affinity analysis systems using collaborative filtering and association rule mining (Apriori, FP-Growth) to deliver a 15% lift in personalized customer engagement strategies.
- Optimized data ingestion throughput by 35% by designing and integrating Java-based microservices with Kafka and Hive for high-velocity data processing.
- Orchestrated scalable end-to-end ML pipelines within Hadoop and Apache Spark ecosystems, ensuring 99.9% reliability for both batch processing and near real-time analytics use cases.
- Integrated streaming and batch data pipelines using Kafka, Spark Streaming, Hive, and SQL-based data stores, enabling timely insights and model-driven decision support.
- Reduced batch processing time by 50% by successfully migrating legacy monolithic ML workflows to a distributed Spark-based architecture hosted on AWS EC2.
- Implemented model lifecycle management practices including experiment tracking, versioning, monitoring, and retraining using MLflow, Jenkins, and Apache Airflow, supporting stable production deployments.
- Partnered with cross-functional product managers and data engineers to translate complex business requirements into production-ready AI/ML solutions that met rigorous enterprise security and scalability standards.

EDUCATION:

- Master of Science in Computer Science, George Mason University, Fairfax, VA
- Bachelor of Engineering in Computer Science, Amrita Vishwa Vidyapeetham, Banglore